HANDOUT OF PHYSICAL CHEMISTRY
COURSE : VIKAAS (JA)
TOPIC : NUCLEAR CHEMISTRY
TARGET: JEE (ADVANCED)

Nuclear Stability

Presently 117 elements are known and the total no. of known isotopes are more than1800. Among these isotopes only about 275 isotopes have stable nucleus and all other are radioactive.

Nuclides can be grouped on the basis of nuclear stability, i.e., stable and unstable nucleus.

The stability of any nucleus against protonic repulsions may be explained by the following theories.

1. Mass defect and binding energy

The stability of nucleus has been explained in terms of strong interaction forces known as nuclear forces which operate between sub-atomic particles when they approach each other with in the range of 10^{-12} cm. In nucleus, the particles are closer than 10^{-13} cm (or 1 fermi) and thus, nuclear forces operate. Nuclear forces are basically attractive forces and are responsible for keeping nucleons bound in the nucleus.

- As the distance of separation increases (> 10^{-13} cm) nuclear forces becomes insignificant. The range up to which the nuclear force is effective is called nuclear range. At the distance of separation of 10^{-12} cm, interaction among nucleons becomes electromagnetic.
- Nuclear forces are about 50-60 times more stronger (in nuclear range) than electromagnetic forces.
- These forces are charge independent and operate between p-p, n-n and p-n.

As a result of these foces, the potential energy barrier of nucleus is lowered due to evolution of energy involving mass decay according to Einstein mass-energy relationship, $E = mc^2$. The resultant nucleus thus, acquired lower energy level and attains stability. The mass difference between a nucleus and its constituent nucleons responsible for binding energy, is called the mass defect.

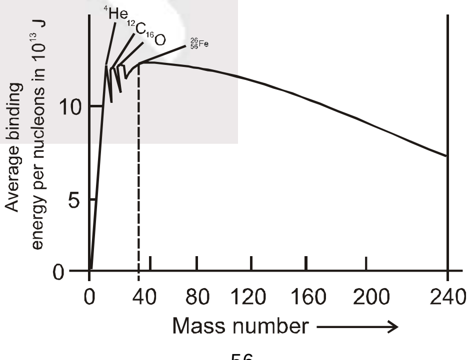
The total energy given out during binding up of nucleons in nucleus is known as binding energy.

Therefore, B.E. = $\Delta m \times c^2$ (Δm = Mass defect)

If Δm = 1 amu then, B.E. = 931.478 MeV, i.e., decay of 1 amu produces 931.478 MeV energy.

B.E. per nucleon, $\bar{B} = \dfrac{\text{Total B.E.}}{\text{No. of nucleons}}$

B.E./nucleon has been found to increase with increase in atomic number and becomes maximum for $_{26}Fe$ at 8.78 MeV. After Fe, it continuously decreases and becomes almost constant at 7.6 MeV for $_{82}Pb$ and onwards.



Note : (1) The greater the binding energy per nucleon, the more stable is nucleus and thus $^{56}_{26}Fe$ is most stable nucleus.

(2) Elements with mass number 4, 12 and 16 have high values of and $\bar{B}$ therefore, $^{4}_{2}He$, $^{12}_{6}C$ and $^{16}_{8}O$ are stable nuclei.

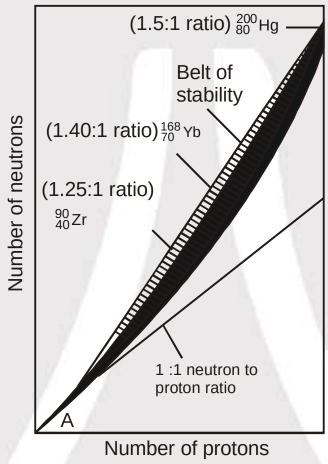
2. The neutron to proton ratio (n/p)

Neutrons apparently help to hold protons together within the nucleus. The number of neutrons necessary to create a stable nucleus increases rapidly as the number of protons increases; the number of neutron to proton ratio of stable nuclei increases with increasing atomic number. The area within which all stable nuclei are found is known as the belt **of stability**. The majority of radioactive nuclei occur outside this belt.

Note : (1) For elements mass number A ≤ 40, Nature prefers the number of protons and neutrons in the nucleus to be about the same to attain stability.

(2) For elements having A ≥ 40, Nature prefers n > p to attain stability. As the number of protons increases, the coulombic repulsive force increases and so the excess of neutrons which produce only attractive force (nuclear force) is required for stability. It is therefore neutron vs. proton curves departs from n = Z line.

(3) Even in light nuclei n may exceed Z but except (in $_1H$ and ^3_2He) is never smaller e.g., $^{11}_5B$ is stable but $^{11}_6C$ is not.



- The type of radioactive decay that a particular radio isotope will undergo depends to a large extent on its neutron to proton ratio compared to those of near by nuclei that are with in belt of stability.

(a) A nucleus whose high n/p ratio places it above the belt of stability emits a β-particle in order to lower n/p ratio and move towards the belt of stability.

$^1_0n \longrightarrow {}^1_0n + {}^0_{-1}e + \bar{\nu}$ ($\bar{\nu}$ is Anti neutrino)   e.g, $^{82}_{34}Se \longrightarrow {}^{82}_{36}Kr + 2\,{}^0_{-1}e$

(b) A nucleus whose low n/p ratio places it below the belt of stability either emits positrons or undergo electron capture. Both modes of decay decrease the number of protons and increase the number of neutrons in the nucleus and thus, positron emission or electron capture results in an increase in n/p ratio. Positron emission is more common than electron capture among the higher nuclei, however, electron capture becomes increasingly common as nuclear charge increases. In such nuclides, the nucleus captures an electron from the K-shell and thus also known as K-electron capture. The vacancy created in K-shell is filled by electrons from higher energy levels giving rise to characteristic X-rays.

$^1_1p \longrightarrow {}^1_0n + {}^0_{+1}e$ e.g., $^{11}_6C \longrightarrow {}^{11}_5B + {}^0_{+1}e$ (Positron emission)

$^1_{+1}p + {}^0_{-1}e \longrightarrow {}^1_0n$ + X-ray   e.g., $^{133}_{56}Ba + {}^0_{-1}e \longrightarrow {}^{133}_{55}Cs$ + X-ray (Electron capture)

Note : A positron has same mass as electron but carries opposite charge. The positron has a very short life because it is annihilated when it collides with an electron, producing gamma rays. This phenomenon is known as pair annihilation. $^0_{-1}e + {}^0_{+1}e \longrightarrow 2\,{}^0_0\gamma$

Similarly pair production occurs when a photon passes through the matter $h\nu = {}^0_{-1}e + {}^0_{+1}e$

(c) The nuclei lying beyond the upper right edge (i.e., nuclei with atomic number > 83), outside the belt of stability, undergo α-emission. Emission of an α-particle decreasing both the number of protons and neutrons and thereby increasing n/p ratio. e.g., $^{238}_{92}U \longrightarrow ^{234}_{90}Th + ^{4}_{2}He$

3. The magic numbers

The nuclear shell model of nuclear structure is analogous to the electron shell model of atomic structure. Just as certain number of electrons (2, 8, 18, 36, 54 and 86) correspond to stable closed shell electron configuration, certain number of nucleons leads to closed shell in nuclei. The protons and neutrons can achieve a closed shell. Nuclei with 2, 8, 20, 28, 50 or 82 protons or 2, 8, 20, 28, 50, 82 or 126 neutrons correspond to closed nuclear shell. Closed shell nuclei are more stable than those that do not have closed shells. These number of nucleons that corresponds to closed nuclear shells are called magic numbers.

Radioactivity

- Radioactivity is a process in which nuclei of certain elements undergo spontaneous disintegration without excitation by any external source.
- The Property of disintegration of radioactive material is independant of temperature, pressure and other external conditions.
- Radioactivity is a nuclear phenomenon i.e. the kind of intensity of the radiations emitted by any radioactive substance is absolutely the same whether the element is present as such or in one of its compounds

Rutherford identified two types of these penetraining rays and named them alpha (α) and beta (β) particles. Later on P. Villard identified and named, the third category as gamma (γ) rays.

Properties of α, β-particles and γ-rays, Becquerel radiations.

S.No.	Properties	Alpha	Beta	Gamma
1	Nature	Fast moving He nuclei	First moving electrons	High energy radiations
2	Notation	$_2He^4$ or α	$_{-1}e^0$ or β	γ or $^0{}_0\gamma$
3	Charge	2 unit (+ ve)	1 unit (–ve)	No charge
4	Typical source	Ra-226	C-14	Tc-99m
5	Velocity	1/10 of light	33% to 90% of light	Same as light waves
6	Nature of path	Straight line	Crooked	Waves
7	Relative penetrating power	1 or (0.01 mm of Al foil)	100 or (0.1 cm of Al foil)	10,000 or (8 m lead or 25 cm steel)
8	Travel distance in air	2-4 cm	200-300 cm	500 m
9	Tissue depth	0.05 mm	4-5 mm	50 cm or more
10	Shielding	Paper, clothing	Heavy clothing, lab coats, gloves	Lead, thick concrete
11	Mass g/particle	6.65×10^{-24}	9.11×10^{-28}	0
12	Relative ionizing power	10,000	100	1
13	Electrical or magnetic field's influence	Deflected towards –ve pole	Deflected towards +ve pole	Not deflected

Theory of Nuclear Disintegration

Rutherford and Soddy

The ejection of α, β and γ-rays from a radioactive material has been satisfactorily explained by Rutherford and Soddy.

Step I : An excited nucleus (a nucleus of low B.E. or higher energy level) tries to attain lower energy level in order to attain stability. Therefore, α-particles come out of nucleus as energy carrier.

Step II : During α-decay no doubt that energy level comes down but n/p ratio increases and therefore to decrease n/p ratio and attain stability, nucleus undergoes neutron decay or neutron transformation to show emission of β-particles. $\quad {}_{0}^{1}n \longrightarrow {}_{+1}^{1}p + {}_{-1}^{1}e + \bar{\nu}$

Step III : The resultant nucleus after α, β-emission still possesses higher energy level than required for its stability, the difference of energy comes out in form of γ-rays. Thus gamma radiations are given off by nuclei in an excited state.

Therefore, α, β-emission are primary emissions and γ-emissions are secondary emissions.

Note : (1) β-particles originates in the nucleus; they are not orbital electrons.

(2) γ-radiations always accompany α or β-emissions and thus, are emitted after α and β-decay.

Soddy-Fajan's Rule or Group Displacement law

(i) A radiactive atom on losing an α-particle shows a loss in mass number by 4 units and a loss in atomic number by 2 units.

Thus, newly formed element occupies two position left to the parent element in periodic table.

$$\underset{\text{II gp}}{{}_{88}^{226}\text{Ra}} \xrightarrow{-\alpha} \underset{\text{0 gp}}{{}_{86}^{222}\text{Rn}} \xrightarrow{-\alpha} \underset{\text{VI gp}}{{}_{84}^{218}\text{Po}}$$

(ii) A radioactive atom on decay of a β-particles shows a gain in its atomic number by 1 unit whereas, mass number remains same.

Thus, newly formed element occupied one position right to the parent element in periodic table.

$$\underset{\text{IV gp}}{{}_{82}^{214}\text{Pb}} \xrightarrow{-\beta} \underset{\text{V gp}}{{}_{83}^{214}\text{Bi}} \xrightarrow{-\beta} \underset{\text{VI gp}}{{}_{84}^{214}\text{Po}}$$

Radioactive Series

A series of nuclear reactions that begins with an unstable nucleus and terminates with a stable one is known as radioactive series or a nuclear disintegration series.

	4*n*	**4*n* + 1**	**4*n* + 2**	**4*n* + 3**
Parent element	${}_{90}Th^{232}$	${}_{94}Pu^{241}$	${}_{92}U^{238}$	${}_{92}U^{235}$
Name of series	**Thorium (Natural)**	**Neptunium (Artificial)**	**Uranium (Natural)**	**Actinium (Natural)**
End product	${}_{82}Pb^{208}$	${}_{83}Bi^{209}$	${}_{82}Pb^{206}$	${}_{82}Pb^{207}$
Number of lost particles	$\alpha = 6$ $\beta = 4$	$\alpha = 8$ $\beta = 5$	$\alpha = 8$ $\beta = 6$	$\alpha = 7$ $\beta = 4$

Nuclear Reactions

The phenomenon of interaction of nucleons giving rise to the formation of a new nucleus or a process in which one nuclide in converted to another by interaction with another nuclide. The **first ever nuclear reaction** in laboratory was carried out by Rutherford when the bombarded N atoms with α-particles.

$${}_{7}^{14}N + {}_{2}^{4}He \longrightarrow {}_{8}^{17}O + {}_{1}^{1}H + Q$$

Another, method of representing this nuclear reaction is ${}_{7}^{14}N\,(\alpha : p)$.

Nuclear Fission :

(a) The phenomenon of splitting up of a heavy nucleus, on bombardment with slow speed neutrons, into two fragments of comparable mass, with the release of two or more fast moving neutrons and a large amount of energy, is known as nuclear fission.

$${}_{92}U^{235} + {}_{0}n^{1} \rightarrow {}_{56}Ba^{140} + {}_{36}Kr^{93} + 3\,{}_{0}n^{1}$$

(b) A loss in mass occurs releasing a vast quantity of energy $\approx 2.041 \times 10^{10}$ kJ per mol of ^{235}U

Nuclear Fusion :

(a) The phenomenon of joining up of two light nuclei into a heavier nucleus is called fusion, e.g.,

$${}_1^2H + {}_1^3H \longrightarrow {}_2^4He + {}_0^1n \; ; \qquad (\Delta H = -17.6 \text{ MeV})$$

(b) Huge amount of energy of required to overpower the Coulombic forces of repulsion in between two nuclie which is obtained by triggering on nuclear fission.

Q.1 In a nuclear transition

$$\underset{\text{(excited state)}}{{}_8^{16}O} \longrightarrow \underset{\text{(ground state)}}{{}_8^{19}O} + 1.06 \times 10^8 \text{ kcal/mol}$$

with what mass per mole the two nuclei differ ?

Q.2 ^{23}Na is the most stable isotope of Na. Find out the process by which ${}_{11}^{24}Na$ can undergo radioactive decay.

Q.3 The number of β-particle emitted during. The change ${}_a^cX \longrightarrow {}_d^bY$ is :

(A) $\frac{a-b}{4}$ (B) $d + \left(\frac{a-b}{2}\right) + c$ (C) $d + \left(\frac{c-b}{2}\right) - a$ (D) $d + \left(\frac{a-b}{2}\right) - c$

Q.4 The decay product of ${}_7^{13}N$ is :

(A) ${}_8^{13}O + {}_{-1}^{0}e$ (B) ${}_6^{13}C + {}_{+1}^{0}e$

(C) ${}_6^{13}C$ + K electron capture (D) ${}_5^9Be + {}_2^4He$

Q.5 A radioactive element X has an atomic numbers of 100. It decays directly into an element Y which decays directly into an element Z. In both processes a charged particle is emitted. Which of the following statement would be true ?

(A) Y has an atomic number of 102. (B) Z has an atomic number of 101.

(C) Z has an atomic number of 97. (D) Z has an atomic number of 99.

ANSWER KEY

1. $\Delta m \approx 5 \times 10^{-3}$ g. **2.** β-emission. **3.** (C)

4. (B) **5.** (D)